

Table 1. Correlations of Nu and Sh numbers over the past 30 years.

Correlation no.	Correlation	Reference
1	$Nu = 3.66(1 + 0.095 \frac{D}{L} RePr)^{0.45}$ $Sh = 3.66(1 + 0.095 \frac{D}{L} ReSc)^{0.45}$	[17]
2	$Nu = 0.571(Re \frac{D}{L})^{\frac{2}{3}}$ $Sh = 0.705(Re \frac{D}{L})^{0.43} Sc^{0.56}$	[19]
3	$Nu = 0.0767(1 + \frac{D}{L} RePr)^{0.829}$ $Sh = 0.0767(1 + \frac{D}{L} ReSc)^{0.829}$	[20]
4	$Sh = 0.766(\frac{D}{L} ReSc)^{0.483}$	[21]
5	$Nu = A[1 + B(Gz)^n \exp(-\frac{C}{Gz})]$	[17, 22, 23]
6	$Nu = aRe^b Pr^c$	[24] [18]

2. Methodology

This section comprehensively covers the experimental setup, procedural details, and mathematical models employed for validation. Initially, it outlines the experimental system and requisite procedures. Following this, it enumerates the governing equations for the energy and momentum transfer, alongside their associated boundary conditions and simplifications for a 0D (lumped), 1D, and 3D computational models. The method of lines (MOL) and finite differences method (FDM) approaches were used to 1D temperature equations within the solid and fluid domains, and the finite element method (FEM) for addressing momentum and energy equations in the 3D model. The discussion extends to encompass parameters, outcomes, and methodologies utilized for

comparing the 0D and 1D models against literature Nu number correlations, and to obtain the appropriate parameters for the given receiver.

2.1. HFSS and Volumetric Receiver Configuration

The experimental set up utilizes artificial light sources with ellipsoidal reflectors to concentrate the light on the frontal opening of the volumetric receiver. The light source and reflector are what constitutes the high flux solar simulator (HFSS) as depicted in **Figure 1B**. The HFSS system includes seven lamps with ellipsoidal aluminum coated reflectors (**Figure 1B**). The lamps were 6 kWe Xe short arc connected to a controller for configuring the positions of the lamps. The lamps were in a squeezed hexagonal configuration with the seventh lamp in the center (**Figure 1B**). For safety purposes and control of temperature, the HFSS is contained in a $7 \times 3.5 \times 2.5 \text{ m}^3$ (LxWxH) compartment made from aluminum composite panels. The reflectors have two focal points; one at 90 mm and another at 2,790 mm from the vertex on the horizontal axis, with a 735 mm opening and 450 mm length. To achieve the maximum flux, the lamp is placed at the first focal point (90 mm from the vertex) and the target object, the volumetric receiver in this case, is placed at the second focal point (2,790 mm from the vertex). More details about the design and operation of the HFSS can be found in the literature [25].

The volumetric receiver used is a commercial square channeled extruded honeycomb made from silicon carbide (SiC) and other additives [Silicon oxide (SiO_2), SiC- Miossanite-6H, and SiC- Miossanite-3C]. It is 137 mm long (L) with a width of 19 mm (w_{total}) and a total of ($n_{channel}$) 100 channels. A channel has width ($w_{channel}$) 1.5 mm and a wall thickness ($t_{channel}$) of 0.4 mm. The whole receiver is placed horizontally facing the HFSS, and enclosed in an aluminum cavity (142 mm in diameter) surrounded with alumina felt insulation ($\sim 0.078 \text{ W m}^{-1} \text{ K}^{-1}$) to thermally insulate the peripheral outer walls of the monolith. **Figure 2** presents a schematic of the volumetric receiver utilized in the experiments, highlighting the key transport phenomena occurring within it. To capture the temperature profiles of the different domains, type-N thermocouples, with Nicrotherm DTM sheath (OD 1.5 mm x 150 mm long, $\sim \pm 2.2 \text{ }^\circ\text{C}$, by TCDirect UK, were placed to on the external walls of the monolith (TI-8, 9, 10). Type-N miniature thermocouples, with Nicrotherm DTM sheath (OD 0.5 mm x 500 mm long, $\sim \pm 2.2 \text{ }^\circ\text{C}$, by TCDirect UK were placed in the internal walls of the monolith (TI-11, 12) temperatures' (**Figure 3**). Two type-N thermocouples were placed to measure the gas inlet (TI-1) and outlet temperature (TI-3) with the same specifications as the external wall thermocouples. The cooling coil and water flanges were placed to cool the air exiting the monolith using type-K with InconelTM 600 sheath (OD 3 mm, IEC60584 Class 1: $\sim \pm 1.0 \text{ }^\circ\text{C}$ by Omega Engineering Inc UK) thermocouples (TI-4, 6, 7) and in the alumina felt insulation (TI-2) to monitor any temperature change.

The monitoring, control, and operation of the HFSS was carried out using LabView environment by National Instruments. The interface of LabView provided the outputs of the thermocouples in the form of a digital signal as temperature. A temperature input module NI 9213 by National Instruments was used to collect the voltage output as an analog signal with a 1 Hz sampling rate, which is then converted to a digital signal with an error less than 0.25°C .

Four volumetric flow controllers (type GFC 17 by AALBORG Instruments & Controls, Inc US) in the range of $0 - 5 \text{ standard L min}^{-1}$ (0.02 L min^{-1} precision, accuracy $\pm 1.0 \%$ of the full scale) were used to control the air flowrate. The flow controllers are four instead of one to improve the accuracy to 0.05 L.min^{-1} instead of 0.2 L.min^{-1} . The differential pressure (PI-1) along the monolith was measured using a differential pressure transducer ($\pm 200 \text{ mbar}$; PD33X, accuracy $\pm 0.1 \%$ of

the full scale, by Keller and purchased from Comer srl, Italy). An analog input module (ADAM-4117 by Advantech,) was used to measure and collect the analog signals from the flow and pressure controllers with 1 Hz sampling rate.

2.2.Measurement of Irradiance

To quantify the flux on the target (volumetric receiver), characterization of the heat flux was conducted with the details provided in [25]. Simply put, a three-step method was used for mapping the irradiance on the target, known as flux characterization. This enables the spatial flux distribution determination with minimum direct measurements and high accuracy. In this scenario, the target is a Lambertian reflector (250x250 mm² stainless steel slab plasma coated with Al₂O₃ made to order by Haueter Engineering, Switzerland) mounted on a 3-axis slider (NLS8–500–101, NLS8–300–102, and NLS8–200–101 precision sliders by NewMark) with a resolution of 0.08 μm and a 1 μm repeatability. The Lambertian reflector enforced the incident light to be reflected in all direction equally and the target was equally bright in all viewing angles [26].

In a similar fashion, a heat flux gauge (HFG) that is a Gardon-type circular foil radiometer (HFG by Vatell: TG1000-0 with a colloidal graphite coating, calibrated with 1% repeatability, 2 mV (W cm⁻¹)⁻¹ sensitivity, and a transducer calibration accuracy of ± 3%) was placed at the Lambertian target center to measure the irradiance at several locations. The output of the HFG (order of 0-10 mV) was recorded using a 24-bit thermocouple input module (NI-9211 by National Instruments) at high resolution and voltage (range ± 80 mV) reading mode. The HFG and Lambertian target were connected to type-K thermocouples (OD 3 mm, IEC60584 Class 1: ~ ± 1.0 °C by Omega Engineering Inc UK) to maintain their temperature at 25°C using a water-cooling system (F50-ME chiller by Julabo).

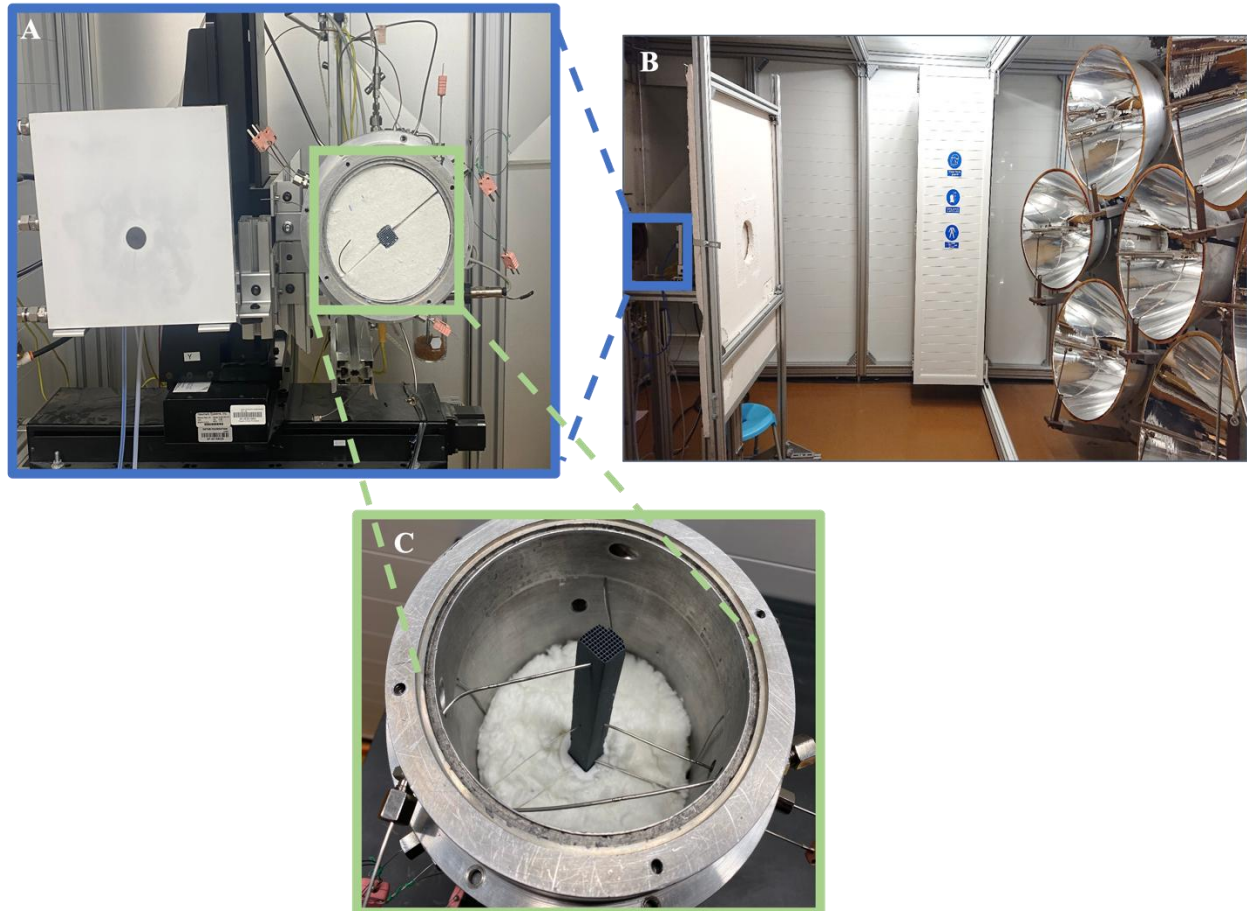


Figure 1. Photos (A) of the assembled volumetric receiver setup and HFG left to the volumetric receiver, (B) the high flux solar simulator, and (C) the receiver during the building stage.

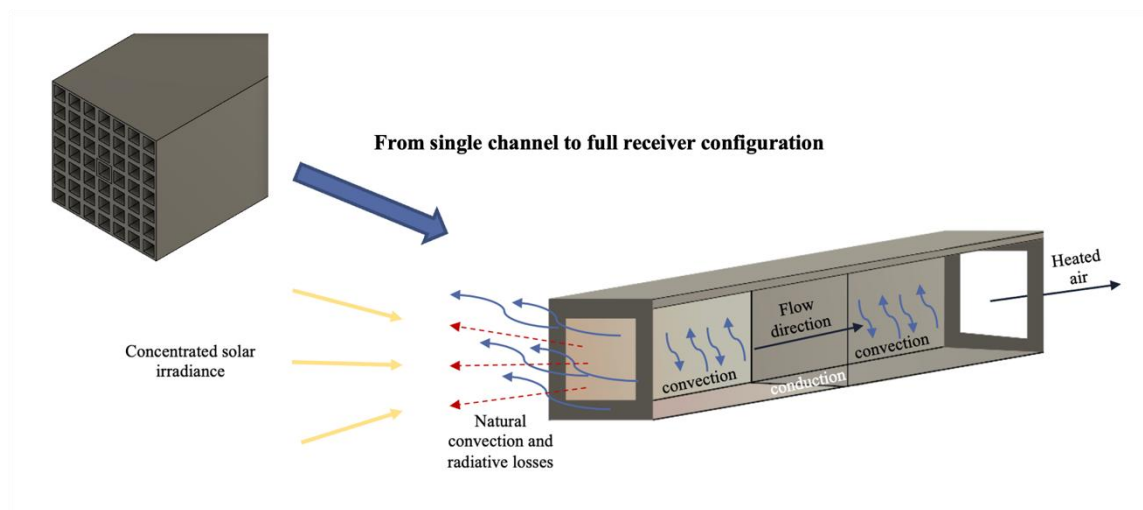


Figure 2. Illustration of the transport phenomena occurring in the volumetric receiver.

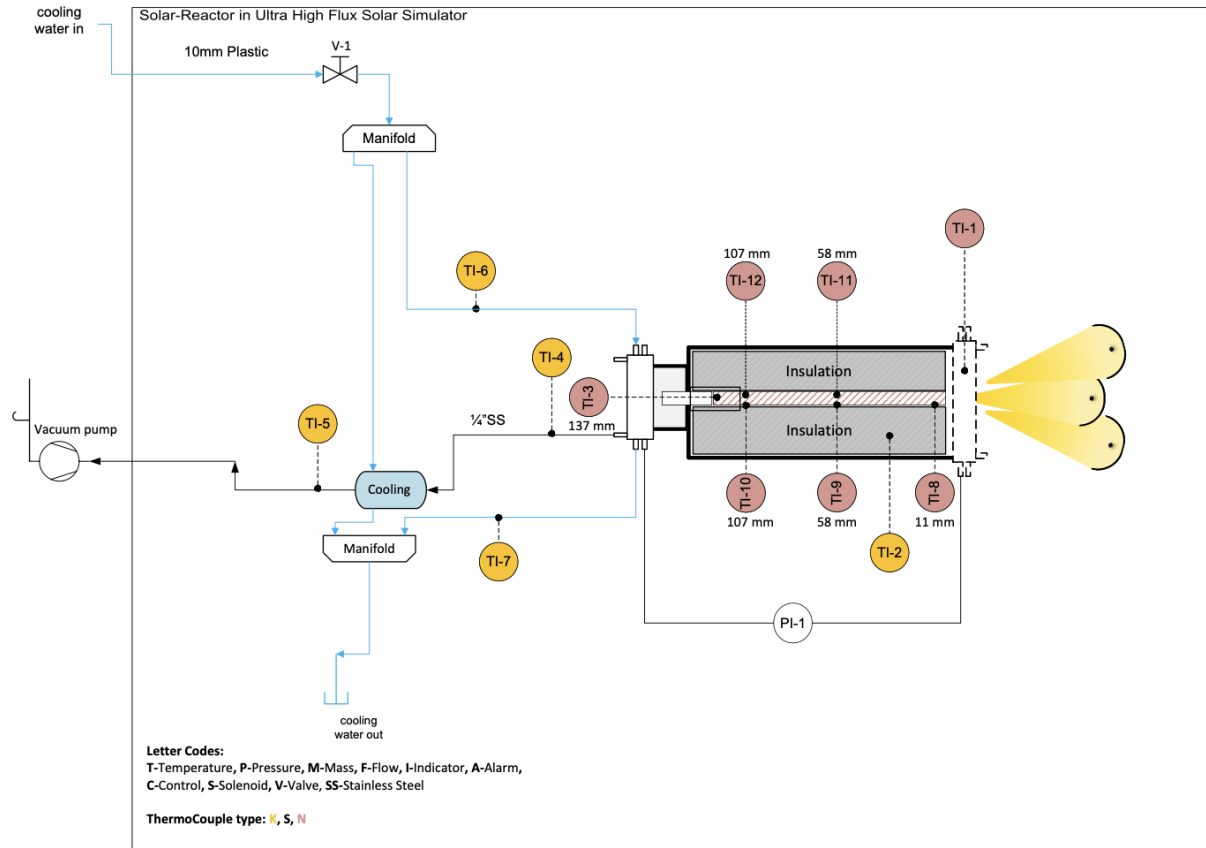


Figure 3. Process and instrumentation diagram (PID) of the experimental solar receiver system.

2.3. Experimental Design and Procedure

The aluminum cavity, with the receiver, is mounted also on the 3-axis slider (**Figure 1A**) with its frontal face facing the HFSS light sources. Ambient air is sucked into the receiver because of a vacuum pump (VACUUBRAND RZ 2.5 rotary vane vacuum pump, 2 stages: maximum pumping speed of 2.3 m³/h at 50 Hz and ultimate vacuum of 2x10⁻³ mbar) connected at the outlet of the mass flow controllers (**Figure 3 - PID**). The experiments were operated at three different (irradiance) powers on the aperture, resulting to several power ratios summarized in **Table 2**. The air outlet temperature and the solid temperatures were measured at different positions from the inlet (see **Figure 3**). The full experimental dataset is available at our GitHub repository [<https://github.com/tamu-edu/tamuq-chen-secarelab-receiver/tree/OneDimension>].

Table 2. Campaign for the volumetric receiver experiments.

Flux (kW/m²)	Flowrate (L/min)	Power ratio (kJ/kg)	Re number per channel	Average solid external temperature (°C)	Average solid internal temperature (°C)	Gas temper -ature (°C)
456	15.3	539	66	649	669	491
	12.5	658	54	695	702	500
	10.5	784	45	717	712	497
	9.12	904	39	771	743	506
	7.1	1156	31	778	728	480
304	18.3	299	79	462	499	380
	13.2	417	57	548	572	421
	9.0	608	39	616	612	425
	7.0	790	30	636	611	408
	4.5	1211	19	641	588	374
256	13.9	334	60	300	328	252
	10.0	461	43	354	376	281
	8.0	575	35	383	396	287
	6.6	698	28	403	408	288
	4.5	1020	19	424	413	277

In **Table 2**, the Re number was calculated to be in the range of 19 to 66, falling within the laminar flow region.

2.4.Numerical Modeling

For the numerical modeling, typical assumptions state that the SiC ceramic material, which serves as the absorber, possesses consistent and uniform properties, exhibits scattering in all directions equally (isotropic scattering), and behaves as a body that emits and absorbs radiation uniformly across all wavelengths (gray emitting and absorbing body). The overall features of the SiC ceramic monolith are determined by its relative density, pore size, thermal conductivity, and mechanical strength [27]. Several models were built to be validated using the experimental data.

Initially, a lumped model was calibrated to match the experimental data by adjusting the thermophysical properties of SiC, the heat transfer coefficient, and the incoming heat flux to improve the accuracy of the temperature profiles as it has few unknowns. Subsequently, these adjusted properties were applied to more complex 1D and 3D models, to further verify the modifications made to the properties of SiC and improve the accuracy of the temperature predictions. This section presents the relevant model equations for the 0D, 1D, and 3D models, along with the initial and boundary conditions applied to each.